



THOMPSON RIVERS UNIVERSITY

Department of Engineering

Safety Critical Software Systems – SENG 4630

Car Autopilot / Driver Assistance System (ADAS)

Term: Winter 2026

Total Marks: 20%

This is a group project.

Project Due: Last day of class - April 14, 2026

Student Learning Outcome:

LO1: Understand the software requirements of developing safety critical systems.

LO2: Apply software life cycle activities into developing critical safety systems.

LO3: Use a programming language to write simple safety critical application.

Degree of Complexity: Students are expected to go above and beyond the direct utilization of algorithms and methods illustrated during the course.

Prerequisites: Software Engineering Design: Process & Principles (SENG-3120)
Algorithms and Data Structure (SENG - 3110)

Project Description:

Modern vehicles increasingly rely on software-controlled driver assistance systems, where failures may result in serious injury or loss of life. In this project, students will design and implement a Simplified Car Autopilot / Driver Assistance System (ADAS) that assists a driver while ensuring fail-safe operation under normal and faulty conditions. The system must safely coordinate between:

- **Vehicle sensors** (speed, distance, lane position – simulated)
- **Control logic** (cruise control, lane keeping, emergency braking)
- **Vehicle actuators** (throttle, braking, steering – simulated)
- **Driver inputs** (engage, disengage, manual override)

The system must be fail-safe, deterministic, and robust against sensor failures.

Students will apply safety-critical design principles, Ada tasking, contracts, and exception handling to meet strict safety requirements.

Key Features:

- Engage and disengage autopilot safely
- Maintain safe vehicle speed (Adaptive Cruise Control – simplified)
- Detect unsafe conditions and apply emergency braking
- Allow immediate driver override
- Prevent unsafe states (e.g., autopilot active with invalid sensors)
- Timeout handling and fault detection for sensor failures

Safety Focus:

- Fail-safe defaults
- Explicit safety rules
- State-based logic
- Exception handling

Ada Concepts Used:

- Tasks and rendezvous
- Protected objects
- Contracts
- Enumerated **types for states**

Project Constraints: Your solution **should consider** if not all but **at least four** from the following list of overarching **characteristics of appropriate design**:

- A. Economic factors
- B. Regulatory compliance (Security and Access)
- C. Reliability
- D. Ethics
- E. Societal impacts
- F. Health and safety
- G. Sustainability

Design: You need to go through the **structured design process**, including the selection of suitable algorithms and data-structures and justifying the reasons behind their selections while satisfying the following requirements:

1. Implementation should be done in Ada 2012 programming language.
2. Execution complexity should be lowest.
3. Failure conditions should be classified based on their severity level and appropriate alerts and measures should take place.

Teamwork: Each group must uphold the best teamwork strategy, including team formation, time management, and conflict resolution (please review ENGR1200 material on teamwork

practices).

Project Deliverables: Following are required to be delivered as the output of the project

1. A fully functional prototype of the alert system (10%)
2. Technical report. (5%)
3. Oral presentation. (15%)

S. No.	Deliverables	Report Components	Timeline
1.	Problem Definition (10%) Design Selection (10%)	Problem definition (Section1) Design Problem (Section 2)	Feb. 25
2.	Iterative Engineering Design Process (20%)	Solution (Section 3)	Mar. 09
3.	Real Implementation (20%)	Solution (Section 3.3)	Mar. 30
4.	Collaborative teamwork and communication skills (10%)	Teamwork and Project Management (Section 4-5)	Apr. 06

- Students are required to fill Teamwork and Project Management section of the report simultaneously with every report component and should be updated regularly.
- After 2nd week, weekly meetings will be held to monitor the progress of the work

Technical Report: The report must follow the provided template. Anyone who reads the report should be able to answer the following questions:

1. Who are possible stakeholders? (Section 1 of the report)
2. What are their needs? (Section 1 of the report)
3. What are the functions, objectives and constraints? (Section 2.2 of the report)